

spaCy, scispaCy, and MedSpaCy

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AI in Health AI 380 H
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Data Preprocessing

1. Investigate ICD-9 diagnosis cohorts. Looking for:
 - a. Reasonable amount of data to work with: $2,000 < \text{num_notes} < 3,000$
 - b. *This rule was derived after many hours of iteration after waiting for my laptop to process large chunks of data. I eventually settled on this range, and later pivoted to work with even smaller samples of the notes due to CPU constraints.*
2. Selected hyperpotassemia - 2,169 discharges.
3. Download notes for this cohort.
 - a. I had to add a limit of 100,000 rows to the query for feasibility on my machine
4. Pickle the notes to a file for usage across all 3 tools

spaCy

spaCy

1. Create a corpus of entities, as shown in the example notebook. I also created an entity dictionary for use as the t-SNE plot legend later.
2. I used a reproducible sample of 20% of the total notes.
3. Pickle the resulting DataFrames.
4. Code and t-SNE plot on following slides.

spaCy NER pipeline

```
PE = "spacy_entities.pkl"
PC = "spacy_corpus_entities.pkl"
SAMPLE_FRAC = 0.2
MAX_CHARS = 20000

if os.path.exists(PE) and os.path.exists(PC):
    spacy_entities = pickle.load(open(PE, "rb"))
    spacy_corpus = pickle.load(open(PC, "rb"))
else:
    nlp = spacy.load("en_core_web_sm", exclude=["parser", "lemmatizer", "attribute_ruler", "tagger"]) # Do not load the parser, lemmatizer, attribute ruler, and tagger

    text = noteevents_df['text'].dropna().sample(frac=SAMPLE_FRAC, random_state=42)
    text = text.str.slice(0, MAX_CHARS)

    spacy_entities = {}
    spacy_corpus = []

    for doc in tqdm(nlp.pipe(text, batch_size=32, n_process=4), total=len(text), desc="Extracting entities"):
        ents = [e.text.lower() for e in doc.ents]
        spacy_corpus.append(ents)

        for entity in doc.ents:
            spacy_entities.setdefault(entity.text.lower(), entity.label_)

    pickle.dump(spacy_entities, open(PE, "wb"))
    pickle.dump(spacy_corpus, open(PC, "wb"))
```

word2vec on spaCy entities

```
"""
Train word2vec on the corpus
"""

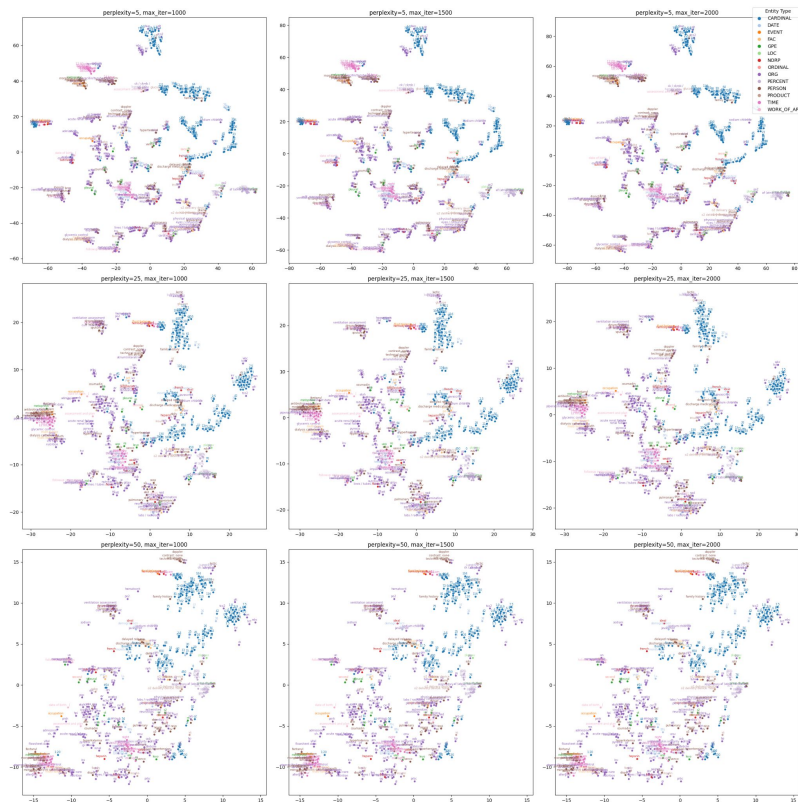
from gensim.models import Word2Vec

# Removing these entities because it filled the plot with numbers
# skip = {"CARDINAL", "PERCENT", "QUANTITY", "TIME", "DATE", "ORDINAL", "MONEY"}
skip = []

corpus_filtered = [
    [e for e in note if spacy_entities.get(e, "") not in skip]
    for note in spacy_corpus
]

model = Word2Vec(corpus_filtered, min_count=5)
```

Tuning spaCy t-SNE plots

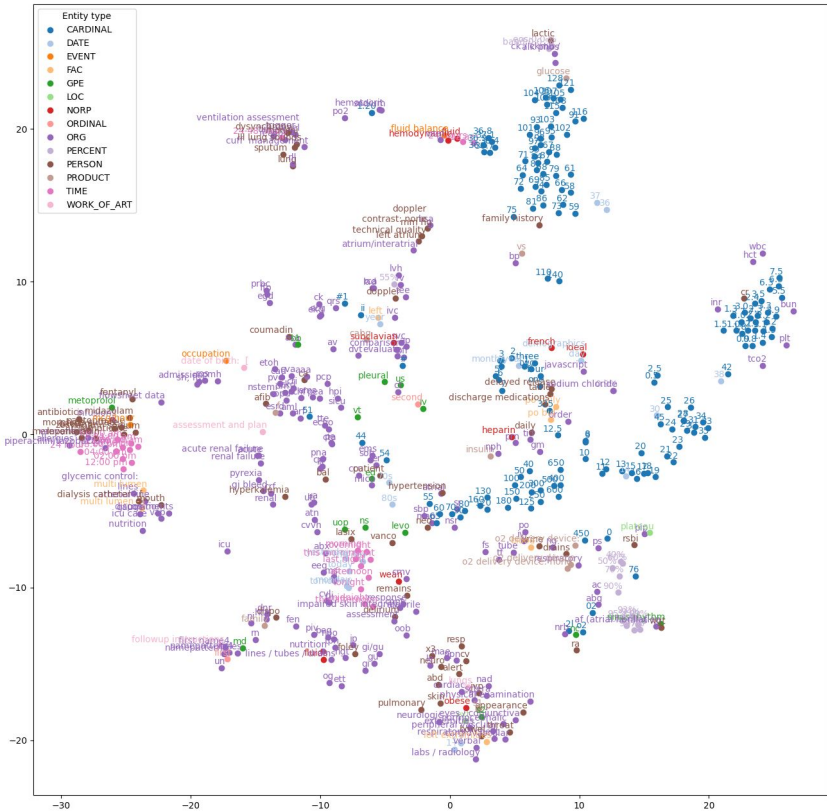


The best tuned plot appears to me to be the plots in the position [1,0:2].

This uses perplexity=25 and max_iter=1000-2000.

Since the plot seems to converge at 1000, there is no need to require more iterations.

Tuned spaCy t-SNE plot



We can see clear groupings surrounding entities, although it is clear that many entities are miscategorized.

spaCy Entity Types

	Mentions	Unique	% mentions
CARDINAL	232509	14360	29.2
ORG	221945	16710	27.8
DATE	104605	19132	13.1
PERSON	73106	7765	9.2
TIME	49292	5185	6.2
GPE	23439	1522	2.9
PERCENT	19588	1714	2.5
QUANTITY	15216	3642	1.9
PRODUCT	12472	1838	1.6
NORP	12029	549	1.5
MONEY	6991	2558	0.9
FAC	6977	905	0.9
ORDINAL	6672	256	0.8
WORK_OF_ART	5959	1281	0.7
LOC	2987	437	0.4
LAW	1907	741	0.2
EVENT	1640	254	0.2
LANGUAGE	149	12	0.0

spaCy did catch a lot of numbers as entities, which polluted the results quite a bit. If I was to run this again, I'd skip the cardinal entity type in the final product.

I also observe spaCy miscategorizing almost everything.

scispaCy

scispaCy

1. Again, create a corpus of entities, as shown in the example notebook. I also created an entity dictionary for use as the t-SNE plot legend later.
2. I used a reproducible sample of 20% of the total notes.
3. Pickle the resulting DataFrames.
4. Code and t-SNE plot on following slides.

scispaCy NER pipeline

```
PE = "scispaCy_entities.pkl"
PC = "scispaCy_corpus_entities.pkl"
SAMPLE_FRAC = 0.2
MAX_CHARS = 20000

if os.path.exists(PE) and os.path.exists(PC):
    scispaCy_entities = pickle.load(open(PE, "rb"))
    scispaCy_corpus = pickle.load(open(PC, "rb"))
else:
    nlp = spacy.load("en_ner_bc5cdr_md", exclude=["parser", "lemmatizer", "attribute_ruler", "tagger"])

    text = noteevents_df['text'].dropna().sample(frac=SAMPLE_FRAC, random_state=42)
    text = text.str.slice(0, MAX_CHARS)

    scispaCy_entities = {}
    scispaCy_corpus = []

    for doc in tqdm(nlp.pipe(text, batch_size=32, n_process=4), total=len(text), desc="Extracting entities"):
        ents = [e.text.lower() for e in doc.ents]
        scispaCy_corpus.append(ents)

        for entity in doc.ents:
            scispaCy_entities.setdefault(entity.text.lower(), entity.label_)

    pickle.dump(scispaCy_entities, open(PE, "wb"))
    pickle.dump(scispaCy_corpus, open(PC, "wb"))
```

word2vec on scispaCy entities

```
"""
Train word2vec on the corpus
"""

from gensim.models import Word2Vec

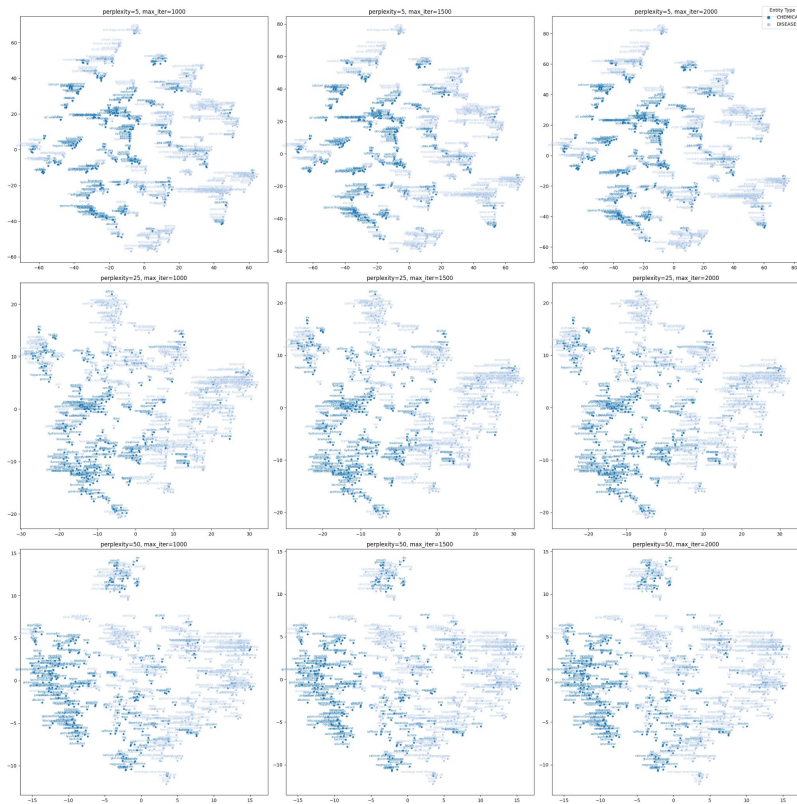
# Removing these entities because it filled the plot with numbers
# skip = {"CARDINAL", "PERCENT", "QUANTITY", "TIME", "DATE", "ORDINAL", "MONEY"}
skip = []

corpus_filtered = [
    [e for e in note if scispaCy_entities.get(e, "") not in skip]
    for note in scispaCy_corpus
]

model = Word2Vec(corpus_filtered, min_count=5)
```

✓ 2.2s

Tuning scispaCy t-SNE plots

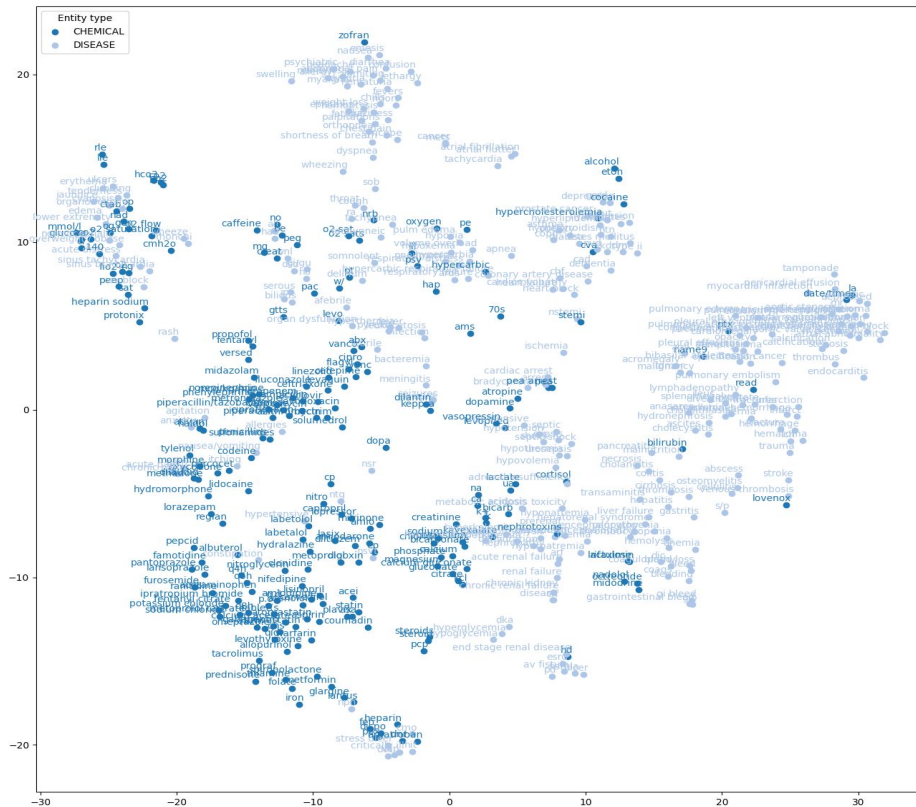


Again, the best tuned plot appears to me to be the plots in the position [1,0:2].

This uses perplexity=25 and max_iter=1000-2000.

Since the plot seems to converge at 1000, there is no need to require more iterations.

Tuned scispaCy t-SNE plot



We can see clear groupings surrounding entities, and it's clear that the entities are correctly categorized this time.

scispaCy Entity Types

This method does not have the same issues as spaCy. Due to its specialized nature, it captures diseases and chemicals accurately.

	Mentions	Unique	% mentions
DISEASE	201972	13064	58.2
CHEMICAL	145294	6594	41.8

MedSpaCy

MedSpaCy

1. Again, create a corpus of entities, as shown in the example notebook. I also created an entity dictionary for use as the t-SNE plot legend later.
2. I used a reproducible sample of **5%** of the total notes.
 - a. **I had to reduce the sample size, this method was far too computationally expensive for my laptop. I understand this has effects on the validity of the conclusions across all three methods, but it was unavoidable in this case.**
3. Pickle the resulting DataFrames.
4. Code and t-SNE plot on following slides.

MedSpaCy NER pipeline

```
PE = "scispacy_entities.pkl"
PC = "scispacy_corpus_entities.pkl"
SAMPLE_FRAC = 0.2
MAX_CHARS = 20000

if os.path.exists(PE) and os.path.exists(PC):
    scispacy_entities = pickle.load(open(PE, "rb"))
    scispacy_corpus = pickle.load(open(PC, "rb"))
else:
    nlp = spacy.load("en_ner_bc5cdr_md", exclude=["parser", "lemmatizer", "attribute_ruler", "tagger"])

    text = noteevents_df['text'].dropna().sample(frac=SAMPLE_FRAC, random_state=42)
    text = text.str.slice(0, MAX_CHARS)

    scispacy_entities = {}
    scispacy_corpus = []

    for doc in tqdm(nlp.pipe(text, batch_size=32, n_process=4), total=len(text), desc="Extracting entities"):
        ents = [e.text.lower() for e in doc.ents]
        scispacy_corpus.append(ents)

        for entity in doc.ents:
            scispacy_entities.setdefault(entity.text.lower(), entity.label_)

    pickle.dump(scispacy_entities, open(PE, "wb"))
    pickle.dump(scispacy_corpus, open(PC, "wb"))
```

word2vec on MedSpaCy entities

```
"""
Train word2vec on the corpus
"""

from gensim.models import Word2Vec

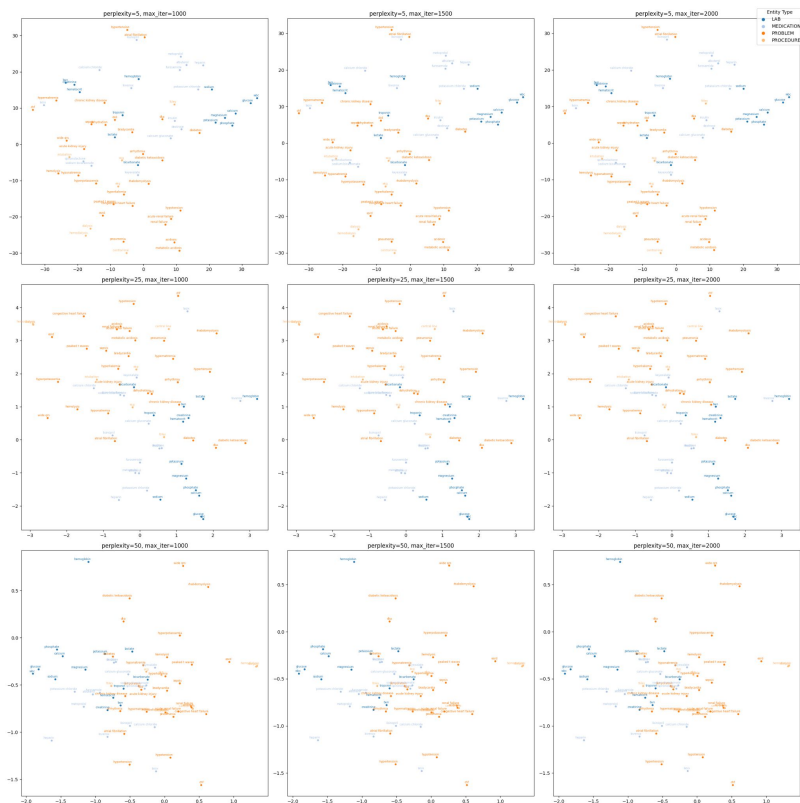
# Removing these entities because it filled the plot with numbers
# skip = {"CARDINAL", "PERCENT", "QUANTITY", "TIME", "DATE", "ORDINAL", "MONEY"}
skip = []

corpus_filtered = [
    [e for e in note if medspacy_entities.get(e, "") not in skip]
    for note in medspacy_corpus
]

model = Word2Vec(corpus_filtered, min_count=5)
```

✓ 0.0s

Tuning MedSpaCy t-SNE plots

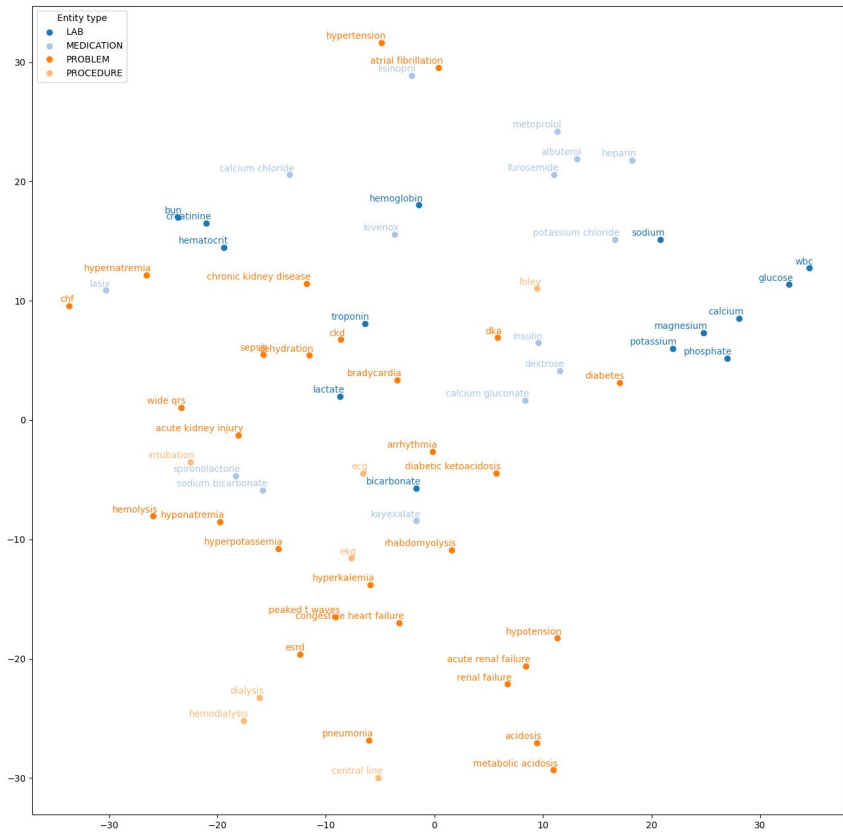


In this case, the perplexity 5 plots look best to me. They accurately group labs and problems the best.

This uses perplexity=5 and max_iter=1000-2000.

Since the plot seems to converge at 1000, there is no need to require more iterations.

Tuned MedSpaCy t-SNE plot



We can see the labs grouped well on the right, as well as related problems grouped near the bottom.

Medications are scattered, but related medications are in the upper right quadrant.

MedSpaCy Entity Types

	Mentions	Unique	% mentions
PROBLEM	7188	30	39.3
LAB	4677	14	25.5
MEDICATION	3943	16	21.5
PROCEDURE	2505	7	13.7

MedSpaCy was accurate, but I did have trouble using it. It was the most confusing of the three methods. I will go into more detail in the comparison slide.

Comparing All 3

Summary per Method

	spaCy	scispaCy	medspaCy
Notes processed	14795.0	14795.0	3699.0
Notes with ≥ 1 entity	14643.0	13948.0	2562.0
Total mentions	797483.0	347266.0	18313.0
Unique entities	78861.0	19658.0	67.0
Mentions / note	53.9	23.5	5.0
Entity types	18.0	2.0	4.0
Appear only once	42926.0	10647.0	0.0
Word2Vec vocab (min_count=5)	15072.0	4145.0	65.0
% appear only once	54.4	54.2	0.0

The reduced sample size used in MedSpaCy is reflected in all categories.

It is clear through several features that scispaCy was a much more focused method than spaCy.

MedSpaCy only captured the ~60 entities for which a rule was created.

Top 15 Entities per Method

	spaCy	scispaCy	medspaCy
1	2 (CARDINAL, 9858)	pain (DISEASE, 6471)	insulin (MEDICATION, 979)
2	1 (CARDINAL, 9323)	pt (CHEMICAL, 3544)	wbc (LAB, 911)
3	response (ORG, 7761)	lasix (CHEMICAL, 3281)	renal failure (PROBLEM, 871)
4	3 (CARDINAL, 6109)	edema (DISEASE, 3260)	hypotension (PROBLEM, 844)
5	icu (ORG, 5479)	heparin (CHEMICAL, 3202)	heparin (MEDICATION, 832)
6	today (DATE, 5087)	hypotension (DISEASE, 3194)	lasix (MEDICATION, 735)
7	5 (CARDINAL, 4694)	allergies (DISEASE, 3046)	glucose (LAB, 735)
8	bp (ORG, 4172)	fio2 (CHEMICAL, 2941)	hypertension (PROBLEM, 654)
9	po (ORG, 4142)	o2 (CHEMICAL, 2929)	pneumonia (PROBLEM, 580)
10	4 (CARDINAL, 3994)	ng (CHEMICAL, 2928)	dialysis (PROCEDURE, 572)
11	first (ORDINAL, 3935)	glucose (CHEMICAL, 2847)	foley (PROCEDURE, 543)
12	abg (ORG, 3725)	arf (DISEASE, 2560)	acute renal failure (PROBLEM, 535)
13	one (CARDINAL, 3700)	dvt (DISEASE, 2559)	hyperkalemia (PROBLEM, 524)
14	wbc (ORG, 3483)	hd (CHEMICAL, 2425)	sodium (LAB, 476)
15	10 (CARDINAL, 3426)	fentanyl (CHEMICAL, 2116)	creatinine (LAB, 462)

It is apparent that spaCy was collecting junk as entities, while scispaCy and MedSpaCy collected meaningful medical terms. We can also observe that the former overlap in several top terms, such as lasix, hypotension, and heparin, to name a few.

Entity Overlap per Method

Pair	Shared unique entities	% of smaller set shared
spaCy vs scispaCy	3260	16.6
spaCy vs medspaCy	61	91.0
scispaCy vs medspaCy	52	77.6

It is interesting that spaCy and scispaCy only shared 16% of their entity set. I believe this is due to spaCy collecting a bloated set of entities unrelated to the medical field, while scispaCy is more medically focused.

MedSpaCy I expected to share almost all of its set, since it was extremely narrowly focused.

Word2Vec Most Similar by Method

```
'potassium'
spaCy : assessment and plan
       fever (0.92), alcoholic (0.92), hyperpotassemia (0.92), chronic (0.91), assessment and plan
       atrial fibrillation (0.91)
scispaCy : bicarbonate (0.92), hyperpotassemia (0.86), calcium (0.86), kayexalate (0.84), sodium (0.81)
medspaCy : magnesium (1.00), bicarbonate (1.00), creatinine (1.00), calcium (1.00), dextrose (1.00)

'hyperkalemia'
spaCy : crf (0.86), mg (0.85), anemia (0.85), rf (0.84), cocaine (0.84)
scispaCy : hyperkalemic (0.89), kayexalate (0.86), nephrotoxins (0.80), hyperpotassemia (0.80), dehydration (0.79)
medspaCy : kayexalate (1.00), peaked t waves (1.00), sodium bicarbonate (1.00), bicarbonate (1.00), ecg (1.00)

'hyperpotassemia'
spaCy : chronic (0.97), control (0.96), hyperthermia (0.96), fuo (0.95), renal failure (0.95)
scispaCy : dextrose (0.95), kayexalate (0.92), insulin/glucose (0.91), ca gluconate (0.90), hyperkalemic (0.87)
medspaCy : hyperkalemia (1.00), bicarbonate (1.00), kayexalate (1.00), sodium bicarbonate (1.00), potassium (1.00)

...

'insulin'
spaCy : ss (0.87), nph (0.86), replete (0.83), sc (0.83), bg (0.82)
scispaCy : hypocalcemia (0.87), insulin/glucose (0.87), renal recs (0.87), citrate toxicity (0.85), nph (0.85)
medspaCy : calcium gluconate (1.00), dehydration (1.00), dka (1.00), spironolactone (1.00), dextrose (1.00)

...

'heparin'
spaCy : haldol (0.87), argatroban (0.87), prn (0.86), tid (0.83), ciwa (0.83)
scispaCy : argatroban (0.75), hsq (0.73), dnr (0.73), dispo (0.73), stress ulcer (0.70)
medspaCy : albuterol (1.00), dextrose (1.00), furosemide (1.00), spironolactone (1.00), atrial fibrillation (1.00)
```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#)...

It is clear to me that scispaCy was the best method at evaluating the notes. The similarity scores were most accurate for scispaCY.

spaCy did not do awful, but it did not excel either.

MedSpaCy did pretty well considering my inexperience in setting up and using it.